

# CHINMAY MURTHY

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ECE graduate specializing in **autonomy, sensor fusion, and real-time systems**. Proven leader in hardware/software integration, delivering high-performance algorithms in C++/Python for industry and research. Passionate about owning complex technical challenges from embedded edge devices to full-stack fleet management.

## EDUCATION

### University of Washington

Seattle, WA

*Bachelor of Science, Electrical and Computer Engineering (3.73) | Music Minor*

Sep 2022 – Jun 2026

- Relevant Coursework: Learning-Based Control, Machine Learning, Computer Architecture, Embedded Systems, Signal Processing, Data Structures & Algorithms, Control Systems

## EXPERIENCE

### Robot Controls Software Lead

Sep 2022 – July 2026

*Advanced Robotics at the University of Washington (ARUW)*

Seattle, WA

- Designed and implemented **Cascade PID-LQG controller** of a **6DOF balancing two-wheel-legged robot** (5-bar active suspension), **simulated in Matlab/Simulink** before **STM32** deployment
- Used **Reinforcement Learning** to train a balancing policy for the same robot with **apprenticeship learning**
- Implemented **inverse kinematics** and gravity feedforward for **mobile 6DOF differential manipulator**
- Designed a **Extended Kalman Filter** for wheel-CV-fused localization achieving **2-3 cm accuracy** over 100m
- Engineered an **autonomous path planner/follower** that can navigate **dynamic obstacles** using **LiDAR**
- Prototyped a **Meta-Imitation Learning** pipeline in **PyTorch** and **Mujoco** distilling **RL and MPC** controllers into a **recurrent foundation policy** for **online system identification** and **zero-shot adaptation**
- Managed **team of 10** to maintain and deploy an **open-source 75k+ LoC C/C++** codebase

### Software Engineering Intern

June 2025 – Sep 2025

*Bidgely*

Palo Alto, CA

- Performed **supervised fine-tuning** of a large language model (**LLM**) on a curated **domain-specific dataset**
- Designed and implemented a **custom retrieval-augmented generation (RAG)** pipeline for proprietary data, reducing **response latency by 47%**

### Undergraduate Research Assistant

Oct 2024 – May 2025

*University of Washington*

Seattle, WA

- Optimized real-time **vision-SLAM and sensor fusion** algorithms (MSCKF-based VINS) leveraging computer vision, IMU, and GPS data for **autonomous robot navigation** in unmapped environments

### Geegah LLC

Ithaca, NY

### Engineering Intern

June 2024 – Sep 2024

- Brought up and verified **RF circuitry** in next-gen (4x pixel density) ultrasonic imager with 200+ components
- Reengineered **FPGA** architecture for **25% lower resource use** and **100x performance gain**
- Created Python API for USB communication, enabling successful **customer demos** and **IEEE IUS** presentation

### Applications Intern

June 2023 – Sep 2023

- Integrated proprietary imager with 3axis motion control stage presented at **DARPA ERI Summit 2023**

## PROJECTS

### Optical Triangulation of Ionospheric Events | *OpenCV, Python*

May 2026

- Meteor detection algorithm and trajectory reconstruction pipeline capable of fusing multimodal data. Developed a custom end-to-end simulation environment for accuracy validation

### Control Tower (Fleet Management System) | *React, Python, GitLab, sCons*

Apr 2026

- Real-time **fleet management dashboard** processing **500Hz telemetry** and aggregating live network metadata from distributed **Raspberry Pi** and **Nvidia Jetson** compute nodes
- **Dynamic IP resolution** database and **API**, integrating with **sCons** to enable **automated firmware flashing** to microcontrollers based on real-time hardware target data

## SKILLS

**Languages:** C/C++, Python, Java, Verilog/SystemVerilog, Matlab, TypeScript

**Tools:** Git, React, MERN, OpenCV, Matlab/Simulink, Unix/Linux, ROS, CAN Bus, Logic Analyzer